

DEVA ANAND

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Education

MS, Computer Science, University of Massachusetts Amherst 2025–2027
Advanced NLP, Advanced Machine Learning, Optimization in CS, Systems for Data Science, Performance Engineering

BE, Computer Science and Engineering, Vels Institute of Science, Technology and Advanced Studies 2019–2023

Professional Experience

Culture & Morality Lab, UMass Amherst, Lab Assistant | *Amherst, MA* Jul 2026 – Present

- Sole engineer building the lab's text-analysis platform (Python, **FastAPI**, sentence-transformers, React) end to end, working independently to own the architecture and every technical decision delegated by the PI.
- Built the platform's engineering backbone: a YAML model registry as single source of truth, a versioned construct library (**99 constructs from 38 questionnaires**), and structured data-quality warnings, growing the backend to **40 hermetic CI tests** that run with no ML dependencies.

Wysa, Backend Engineer | *Bengaluru, India* Jul 2023 – Jul 2025

- Engineered a multi-tenant NHS eTriage backend (Node.js, MongoDB) for **8+ UK clinical clients**, processing **10,000+ monthly triage submissions** via Mayden iaptus REST integrations; contributed to **\$340K+** in revenue.
- Diagnosed and resolved production issues across tenants, monitoring logs during releases and client integrations and shipping targeted fixes for tenant-specific incidents to keep the multi-tenant service stable in a live environment.
- Owned backend security and reliability for the eTriage stack, implementing **AWS KMS** encryption for PII, hardening authentication and rate limiting, and remediating **20+ VAPT findings** with regression-tested fixes.

Cario Growth Services, Backend Developer Intern | *Chennai, India* Nov 2022 – May 2023

- Designed and implemented **20+ RESTful APIs** (Node.js + Fastify, PostgreSQL) with input validation, clear contracts, and backward-compatible changes used in production by the frontend team.
- Migrated the core codebase from JavaScript to TypeScript across multiple repositories, reading through existing code to reason about behavior and improve type safety and long-term maintainability.

Projects

NEON Quantized GEMM – INT8/INT4 SIMD Kernels (GitHub) C++17, ARM NEON, INT8/INT4 quantization, roofline

- Built INT8/INT4 GEMM kernels in C++ for the ARM NEON `sdot` instruction, taking a naive fp32 baseline (3 GFLOP/s) to a 4x4 register-blocked int8 micro-kernel at **394 GFLOP/s** (132x over naive, ~2.5x over the compiler's auto-vectorized code) by restructuring memory layout and hiding instruction latency with multiple accumulators; verified bit-exact against a scalar reference and benchmarked with p50/p99/p999 tail-latency methodology.

Spark ETL Performance Optimization (GitHub) PySpark, Spark UI, SQL, NYC TLC Dataset

- Cut end-to-end ETL runtime ~**25%** on a ~**1.4B-row** dataset and reduced the analytics tail-latency ratio (P99/P50) from **2.14x to 1.74x** via broadcast joins (shuffle minutes to ~0.05s), AQE, skew-join handling, and partition/column pruning, using Spark UI stage metrics to localize bottlenecks.

Sonare – Offline Sign / Speech App (GitHub) React, Electron, FastAPI, MediaPipe, whisper.cpp

- Built a cross-platform desktop app running a fully offline on-device pipeline (MediaPipe hand tracking, whisper.cpp ASR, local TTS) with a FastAPI backend and async queues that keep the UI responsive under real-time video and audio, with no external network dependencies.

Publications

- "Alzheimer's Disease Classification using Transfer Learning," IEEE CONIT 2023, first author, deep transfer learning on neuroimaging data. (Paper)

Skills

Languages & Databases: C++, Python, TypeScript, JavaScript, Java, SQL, Bash, PostgreSQL, MongoDB, SQLite

Systems & Perf: Linux, perf / Cachegrind profiling, SIMD intrinsics (AVX/NEON), roofline analysis, cache-aware tiling, multithreading, Spark / PySpark

Backend & APIs: Node.js, Express, FastAPI, REST APIs, async services, integration testing, Next.js, Docker

ML & AI: PyTorch, NumPy, sentence-transformers, quantization (INT8/INT4), LLM agents, RAG

Cloud & DevOps: Git, GitHub Actions, CI/CD, Docker, distributed systems, AWS, GCP, Azure

Open Source & Community

- Built **Underdogs Fitness** (Website), a production MERN platform with Stripe payments; live **3+ years**. - Selected for **GirlScript Summer of Code 2023**; contributed to Linkfree, Freehit, and ProjectsHut. - Tutored **30+ underprivileged students** in programming at Sayur, a non-profit in Tamil Nadu.